$$N=\frac{V_0}{u}\ln\frac{p_1}{p_a} \tag{20}$$

Equations (17)、(18)、(20) give

$$E_{\mathrm{k}}=(p_1-p_a)V_0-p_aV_0\ln\frac{p_1}{p_a} \tag{21}$$

Substituting (121)(6) into gives

$$E_{\mathrm{k}}=p_av_a\left(1-\frac{T_1}{T_a}+\frac{T_1}{T_a}\ln\frac{T_1}{T_a}\right)。 \tag{22}$$

The electrical energy obtainable from this kinetic energy is

$$E=0.45p_aV_a\left(1-\frac{T_1}{T_a}+\frac{T_1}{T_a}\ln\frac{T_1}{T_a}\right)。 \tag{23}$$

Obtaining E requires the iceberg to absorb $-Q$ of heat. Melting it absorbs a total of

$$Q_1=mL。 \tag{24}$$

Hence the total electrical energy obtainable is

$$E_1=\frac{mL}{-Q}E。 \tag{25}$$

Substituting (5)(22)and into gives

$$E_1=\frac{9}{70}mL\frac{1-\frac{T_1}{T_a}+\frac{T_1}{T_a}\ln\frac{T_1}{T_a}}{1-\frac{T_1}{T_a}}, \tag{26}$$

Substituting the numerical data gives

$$E_1=1.5\times10^{14}\,\mathrm{J} \tag{27}$$

(22)(28)(7)(9)(3) (5) ; ; ; ; ;

3; (25) ~ (27) each .

Method 2:

At ambient pressure p_a, let n moles of air undergo the process as all the ice melts.

In step (a), volume changes by ΔV and temperature by $\Delta T=T_1-T_a$. The air releases the heat

$$Q=-p_a\Delta V-\frac{5}{2}p_a\Delta V。 \tag{1}$$

On the right, term one is atmospheric work on the system; term two is the internal energy change. The equation of state gives